

Air Tracker: Flight Analytics

End-to-End Aviation Data Engineering & Insights

Developed by: Anusha

Project Foundation

Leveraging AeroDataBox API for Strategic Flight Intelligence

Technical Expertise Acquired



Python

Advanced scripting for API consumption, JSON parsing, and data cleaning.



SQL & RDBMS

Relational schema design, normalization, and complex analytical queries.



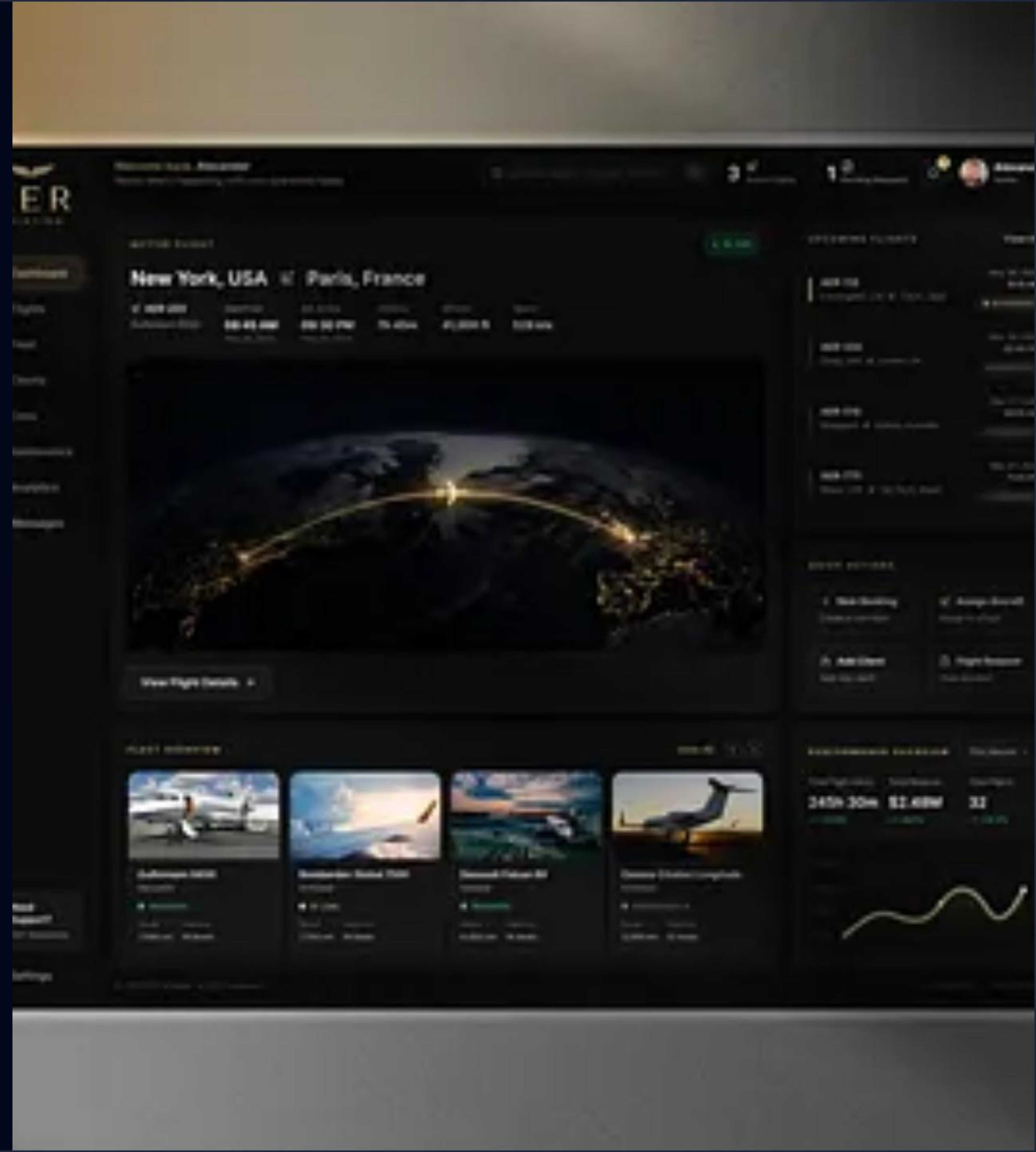
Streamlit

Developing interactive data dashboards for real-time visualization.

The Challenge

Navigating the complexity of aviation data requires a structured approach to manage fragmented JSON responses.

The goal is to analyze operational patterns for **10 key airports**, transforming raw API streams into a high-performance relational database for deep trend exploration.



Strategic Value Creation



Trend Analysis

Visualize distribution of departures and arrivals by status and airline.



Operational IQ

Analyze flight delays and status patterns across selected global hubs.



Decision Support

Provide data-driven insights for airport resource allocation.

Data Extraction Architecture

- ✓ **Extraction:** Targeted polling of 10 airport codes via AeroDataBox.
- ✓ **Transformation:** Flattening nested JSON into structured SQL records.
- ✓ **Enrichment:** Fetching unique aircraft details using filtered flight data.



Relational Schema Design

ENTITY	PRIMARY KEY	KEY ATTRIBUTES
Airport	airport_id	icao_code, name, city, latitude, longitude, timezone
Aircraft	aircraft_id	registration, model, manufacturer, icao_type_code
Flights	flight_id	flight_number, aircraft_reg, origin, destination, status
Delays	delay_id	iata, delay_date, total_flights, avg_delay_min

Advanced Analytical Queries

Operational Volume

Listing outbound flights per airport (threshold > 5) and identifying top 3 destination hubs by arrival frequency.

Efficiency Metrics

Computing delay percentages per destination and categorizing domestic vs. international routes using CASE logic.

Arrival Performance by Airport



On-time arrival distribution across sample airport network.

Development Impact

10

Hub Airports Analyzed

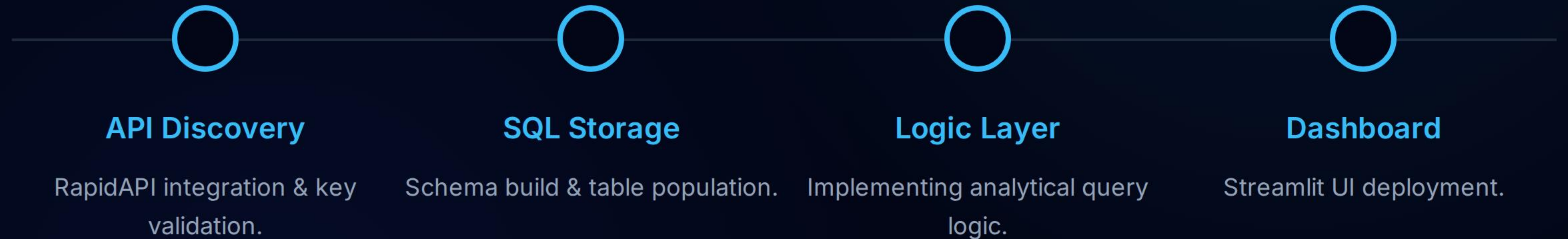
11

Critical SQL Queries

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Unified Dashboard

Execution Roadmap



Questions?

Thank you for your time.

 github.com/anushavenkatesh9535-hub

Image Sources



<https://cdn.dribbble.com/userupload/47518393/file/0f5b964403e59a81875e819f23d1bf24.png?format=webp&resize=400x300&vertical=center>

Source: dribbble.com



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